

LAADAN-AC: A Lagrangian Admissibility-Aware Actor-Critic Proposal

From ICU-Sepsis Benchmarking toward Safe Adaptive Robot Control

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Purpose and Research Motivation

LAADAN-AC was proposed as a safety-oriented offline reinforcement learning model for the ICU-Sepsis benchmark. The immediate purpose was to test whether an actor-critic policy could achieve strong survival-oriented return while avoiding inadmissible actions. The wider research purpose is more ambitious: to investigate whether the same safety structure could later be adapted into a general safe-control framework for real-world and social robotics.

The key idea is that adaptive agents should not only ask, “which action gives the highest reward?” They should also ask, “is this action allowed, safe, expert-supported, and behaviourally smooth?” This follows the constrained Markov decision process view, where a policy maximises return while satisfying a cost or safety constraint (Altman, 1999; Achiam et al., 2017).

Proposed Architecture

LAADAN-AC uses an actor-critic architecture with four main components:

- an **actor** that outputs action logits;
- **twin reward critics** Q_1, Q_2 that estimate long-term reward value;
- a **cost critic** C_ψ that estimates unsafe or inadmissible cost;
- an **admissibility mask** that blocks actions marked unsafe for a state.

For a state s and action a , the actor produces logits $z_\theta(s, a)$. The proposed LAADAN-AC policy applies a mask $m(s, a)$, where $m(s, a) = 1$ means the action is admissible and $m(s, a) = 0$ means it is blocked:

$$\pi_\theta(a|s) = \frac{\exp(z_\theta(s, a))m(s, a)}{\sum_{a'} \exp(z_\theta(s, a'))m(s, a')}.$$

This masking step is a project-specific LAADAN-AC design choice. It gives the direct action-level safety filter: the final policy can only choose from admissible actions. The critics then learn reward and cost estimates over the benchmark transition model.

Constrained Objective and Lagrangian Term

The safety problem can be written as:

$$\max_{\pi_\theta} J_R(\pi_\theta) \quad \text{subject to} \quad J_C(\pi_\theta) \leq d.$$

Here, $J_R(\pi_\theta)$ is expected return, $J_C(\pi_\theta)$ is expected unsafe cost, and d is the allowed cost budget. In the coursework implementation, $d = 0$, meaning the policy is encouraged toward zero unsafe cost.

Following constrained reinforcement learning, the constraint is relaxed using a non-negative Lagrange multiplier λ (Altman, 1999; Achiam et al., 2017):

$$\mathcal{L}(\pi_\theta, \lambda) = J_R(\pi_\theta) - \lambda (J_C(\pi_\theta) - d).$$

In implementation, the actor minimises the proposed LAADAN-AC loss:

$$\mathcal{L}_{actor} = \mathbb{E}_s \left[-\mathbb{E}_{a \sim \pi_\theta} [Q(s, a)] - \alpha \mathcal{H}(\pi_\theta) + \lambda \mathbb{E}_{a \sim \pi_\theta} [C_\psi(s, a)] + \beta D_{KL}(\pi_E || \pi_\theta) + \mu \mathbb{E}_{a \sim \pi_\theta} [S(s, a)] \right].$$

The notation is as follows. $Q(s, a) = \min(Q_1, Q_2)(s, a)$ is the conservative reward estimate from the twin critics. $C_\psi(s, a)$ is the learned cost estimate. $\mathcal{H}(\pi_\theta)$ is the policy entropy, inspired by maximum-entropy actor-critic learning (Haarnoja et al., 2018). $D_{KL}(\pi_E || \pi_\theta)$ keeps the policy close to the expert-safe policy π_E . $S(s, a)$ is the project-defined smoothness or action-deviation cost. The coefficients α, β, μ control entropy, expert guidance, and smoothness strength.

The Lagrange multiplier is updated using projected ascent:

$$\lambda \leftarrow \max(0, \lambda + \eta_\lambda(\bar{C} - d)),$$

where η_λ is the Lagrange learning rate and $\bar{C}$ is the current mean expected cost. If expected cost rises above the budget, λ increases, making future unsafe behaviour more expensive. This is the mathematical reason the model is genuinely Lagrangian. However, the hard zero-inadmissibility behaviour mainly comes from the admissibility mask, while the Lagrangian term adds adaptive cost pressure during learning.

Current Use and Future Research Vision

In the current ICU-Sepsis benchmark, the environment is a fixed tabular MDP. This allows reward and cost critics to be trained using Bellman target updates over known transition probabilities. The model is therefore a genuine constrained offline actor-critic prototype, but it should not be described as a clinical deployment system.

The future research direction is to adapt LAADAN-AC beyond ICU-Sepsis into safer adaptive robot control. This vision is inspired by LeCun’s proposal that autonomous intelligent agents should learn predictive world models and plan in latent representation space (LeCun, 2022). In a future JEPA-inspired LAADAN-AC, the fixed benchmark state would be replaced by a learned latent world model. A social robot could predict the likely consequences of its actions, while LAADAN-AC would apply admissibility constraints, Lagrangian cost control, expert guidance, and smoothness penalties before acting.

Recent JEPA-based work such as LeWorldModel suggests that joint-embedding predictive architectures can learn compact latent world models from pixels for control-related prediction and planning (Maes et al., 2026). This makes LAADAN-AC a potential research bridge between offline safe reinforcement learning and future world-model-based adaptive robotics.

References

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